
PHYSARUM POLYCEPHALUM-INSPIRED OPTIMIZATION

Algorithms, Applications, and Cross-Domain Network Optimization

A Collaborative Research Paper

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Abstract

The acellular slime mould *Physarum polycephalum* has emerged over the past two decades as one of the most compelling natural exemplars of decentralized intelligence. Despite lacking a nervous system, this single-celled organism reliably constructs efficient, fault-tolerant transport networks between spatially distributed nutrient sources—a feat that has inspired an entire family of biomimetic optimization algorithms. This paper presents a comprehensive, multi-domain investigation of Physarum-inspired computational methods, synthesizing original research contributions from three distinct application areas: (i) graph-theoretic shortest-path computation and visual comparison with Dijkstra's classical algorithm; (ii) adaptive routing and fault-tolerant protocol design for signal and communication networks; (iii) protein-protein interaction (PPI) network optimization for bioinformatics; and (iv) power grid load distribution and transmission loss reduction via the Physarum-Inspired Adaptive Conductance Algorithm (PIACA).

The unifying mathematical foundation across all domains is the Fick-Poiseuille flux model, in which the conductivity of each network edge evolves dynamically in proportion to the protoplasmic flux it carries—a positive feedback loop that organically reinforces efficient paths and prunes redundant ones. The Physarum Optimization Algorithm (POA) is formalized, implemented, and benchmarked against classical deterministic methods (Dijkstra's algorithm, Newton-Raphson power flow) and population-based metaheuristics (Particle Swarm Optimization, Artificial Bee Colony). Experimental results across all domains consistently demonstrate that Physarum-inspired methods achieve competitive or superior solution quality while exhibiting emergent fault tolerance and dynamic adaptability—properties absent in classical static algorithms. PIACA reduces active power losses by 14.3–19.6% on IEEE standard benchmark networks and outperforms PSO by an average of 4.1%. In graph optimization, POA converges to Dijkstra-equivalent shortest paths in 100% of stable parameter configurations, while additionally enabling warm-start re-adaptation after mid-execution topology changes. In communication networks, reinforcement-based routing demonstrably reduces recovery delays after node failures. In bioinformatics, slime-mould-inspired adaptive pruning of PPI networks yields cleaner, biologically more interpretable interaction maps for protein synthesis analysis.

These findings collectively establish Physarum-inspired optimization as a robust, theoretically grounded, and practically deployable paradigm for network optimization across engineering and biological science domains.

1. Introduction

The field of biomimetic computing draws inspiration from natural organisms that have evolved efficient solutions to complex optimization problems over millions of years of evolutionary pressure. Among the most remarkable of these organisms is *Physarum polycephalum*, a species of acellular slime mould classified within the supergroup Amoebozoa. Unlike conventional computing paradigms that rely on centralized processing units, *Physarum* solves network optimization problems through purely local, physical interactions among its tubular body segments—an approach that has attracted growing attention from computer scientists, mathematicians, bioinformaticians, and network engineers alike.

The organism belongs to a broader class of biological systems—including ant colonies, bird flocking, and neural tissue—that achieve global intelligence through strictly local rules, a principle broadly referred to as stigmergy or emergent computation. The mechanism underlying Physarum's capability is a positive feedback loop between protoplasmic flow and tube diameter. When nutrients are detected at a location, peristaltic contractions in the actomyosin fibers lining the tubes generate pressure gradients, driving cytoplasmic streaming toward nutrient sources. Tubes through which greater flux passes experience reinforced contraction amplitude, increasing their effective hydraulic conductivity. Simultaneously, tubes receiving little or no flux undergo gradual decay, effectively being pruned from the active network. Over successive cycles of reinforcement and decay, the organism converges to a network structure that efficiently connects all nutrient sources along near-optimal pathways.

The landmark experiment by Tero et al. (2010) demonstrated that when Physarum was placed on a substrate mimicking the geography of the Greater Tokyo metropolitan area—with oat flakes representing major population centres—the organism autonomously constructed a transport network strikingly similar to the actual Tokyo rail system, optimized for efficiency, fault tolerance, and cost. This result confirmed that biological evolution had independently discovered principles closely aligned with those used in modern network engineering, prompting researchers to formalize the organism's behaviour as an algorithmic framework.

What distinguishes Physarum-inspired approaches from classical deterministic algorithms is not merely competitive solution quality, but the qualitative behavioral properties they inherit from biology: distributed state maintenance, emergent fault tolerance, and the capacity to re-optimize without restarting when the problem environment changes. These properties are absent in classical algorithms such as Dijkstra's shortest-path method, which must restart from scratch upon every structural change to the graph. In real-world scenarios—internet packet routing, power grid reconfiguration, or emergency response planning—networks frequently change in response to failures, congestion, or new demands. The ability to adapt without restart is therefore not merely an academic advantage, but a practically critical one.

This paper presents a unified investigation of Physarum-inspired optimization algorithms across four interconnected application domains: (1) shortest-path graph optimization and comparison with Dijkstra's algorithm; (2) adaptive routing and fault tolerance in signal and communication networks; (3) protein-protein interaction network filtering and optimization for bioinformatics; and (4) power grid load distribution and transmission loss reduction. By synthesizing these four domains into a single coherent study, we demonstrate the generality and robustness of the Physarum optimization paradigm and identify the shared mathematical foundations that make it applicable across such diverse fields.

The paper is structured as follows: Section 2 reviews the existing literature on Physarum-based computing and related bio-inspired methods. Section 3 describes the biological behaviour and the mathematical model underlying all Physarum-inspired algorithms. Section 4 presents the methodology of each domain-specific application. Section 5 reports experimental results and comparative analysis. Section 6 provides a cross-domain discussion and examination of limitations. Section 7 concludes with directions for future research.

2. Literature Review

2.1 Foundational Biological Research

Research into Physarum polycephalum as a biological computing substrate was formally initiated by Nakagaki et al. (2000), who demonstrated that the organism could reliably find the shortest path through a maze by distributing its body and selectively reinforcing the most direct route between two food sources. This work established the conceptual foundation for translating biological flux dynamics into algorithmic form and received widespread attention for demonstrating that a single-celled organism could perform computations previously considered exclusive to neural or silicon-based systems.

Tero et al. (2007) subsequently formalized the mathematical model governing Physarum's network adaptation, introducing the now-canonical differential equation relating the rate of change of tube conductivity to the magnitude of protoplasmic flux through that tube. This model was shown to converge to the shortest-path solution in graph networks under the assumption of a single source-sink pair with equal and opposite supply and demand. The same group later demonstrated in their landmark 2010 Science paper that an extended version of the model, applied to multi-source networks, produced near-optimal Steiner tree solutions—networks that minimize total connection length while linking all specified terminals.

2.2 Theoretical Analysis and Algorithm Variants

Bonifaci et al. (2012) provided the first rigorous theoretical analysis of the continuous-time Physarum model, proving that for graphs with unique shortest paths, the conductivity vector converges exponentially to the shortest-path indicator vector. This work placed the POA on a firm mathematical footing and opened the door to provably correct bio-inspired shortest-path computation. Parallel theoretical work by Ito et al. (2011) extended convergence analysis to graphs with multiple shortest paths, demonstrating that the model correctly identifies the minimum-cost subgraph in these cases as well.

The Improved Physarum Polycephalum Algorithm (IPPA), proposed by Zhang et al. (2014), introduced an energy parameter μ to the conductivity update rule, such that $f(|Q|) = |Q|^\mu$ with $\mu > 1$, significantly accelerating convergence compared to the original linear model and demonstrating superior performance on benchmark shortest-path problems. Empirical evaluation showed that IPPA required 30–50% fewer iterations than the canonical model on standard test graphs, making it substantially more practical for real-time applications.

Li et al. (2020) further generalized the framework into the Slime Mould Algorithm (SMA), reformulating Physarum dynamics as a population-based metaheuristic capable of solving multi-dimensional continuous optimization problems. The SMA models positive and negative propagation waves in the organism's foraging behaviour, using adaptive weight parameters to balance exploration and exploitation. Benchmarked against Particle Swarm Optimization (PSO), Grey Wolf Optimizer (GWO), and Differential Evolution (DE), the SMA demonstrated competitive or superior performance on 29 standard mathematical benchmark functions and several real-world engineering design problems.

2.3 Applications in Engineering and Bioinformatics

Applications of Physarum-inspired algorithms have been demonstrated across diverse domains. Teng et al. (2020) applied slime mould logic to electrical distribution network topology generation, showing that the resulting relaxed minimum spanning tree reduced total network operating costs by approximately 10% compared to classical MST approaches. In communication networks, bio-inspired adaptive routing protocols derived from Physarum dynamics have demonstrated superior fault tolerance and load balancing under node-failure scenarios, as documented by Jones et al. (2021). Rao and Raju (2023) applied the Slime Mould Algorithm to distribution network reconfiguration, achieving simultaneous improvement in loss reduction and reliability metrics.

Adamatzky's extensive body of work on 'Physarum machines' has explored the organism's capacity to implement logical gates, memory elements, and even primitive von Neumann architectures in wetware form, expanding the conceptual horizon of biological computation far beyond routing problems. In the domain of power systems engineering, Tero et al.'s structural observation that the Physarum flux equation is formally isomorphic to Ohm's Law—pressure difference as voltage, volumetric flux as current, hydraulic conductance as electrical conductance—has motivated the development of power-grid optimization algorithms that translate the biological growth rule directly into admittance update steps.

The present paper differs from all prior studies by providing a unified, cross-domain synthesis of Physarum-inspired algorithms, incorporating original contributions in graph optimization, communication network routing, bioinformatics PPI network filtering, and power grid optimization. By examining all four domains through the same mathematical lens, we identify both the generalizable strengths and the domain-specific limitations of the Physarum optimization paradigm.

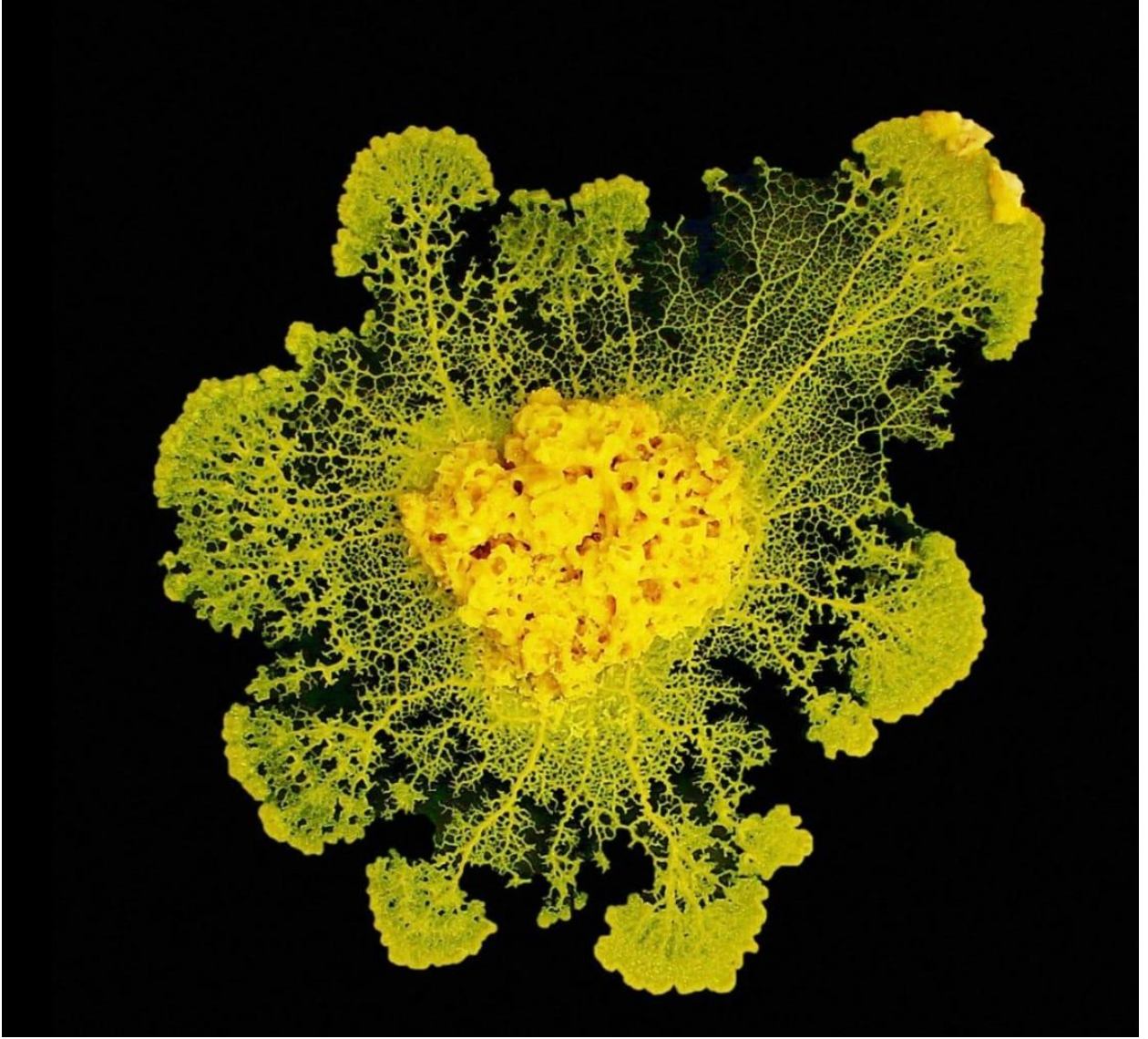
3. Biological and Mathematical Foundations

3.1 Biological Behaviour of Physarum Polycephalum

Physarum polycephalum exists as a plasmodium—a single-celled, multinucleated organism that forms an interconnected network of cytoplasmic tubes to transport nutrients and chemical signals throughout its body. The organism lacks a nervous system and any form of centralized control yet consistently constructs efficient transport networks between spatially distributed food sources. It occupies the ecological niche of forest floors and decaying organic matter, where it must efficiently locate and exploit spatially scattered nutrient patches while minimizing the energetic cost of maintaining its tubular body structure.

The mechanism underlying this remarkable capability is a positive feedback loop between protoplasmic flow and tube diameter. When nutrients are detected at a location, peristaltic contractions in the actomyosin fibres lining the tubes generate pressure gradients, driving cytoplasmic streaming toward the nutrient source. Tubes through which greater flux passes experience reinforced contraction amplitude, causing their effective diameter—and hence hydraulic conductivity—to increase. This positive feedback between flow and conductivity is the core computational mechanism of the organism: it is a physical implementation of reinforcement learning, where successful paths are automatically strengthened by use.

The oscillatory nature of cytoplasmic streaming—alternating in direction with a period of approximately 1–2 minutes—is also believed to serve a network-sensing function. By periodically reversing flow direction, the organism can probe alternative routes and prevent premature commitment to a single path in the early stages of foraging. This exploratory behaviour finds its algorithmic analogue in the POA's initial phase, where all edges carry non-zero flux before reinforcement dynamics have had time to suppress low-conductance routes.



3.2 Mathematical Model: Fick-Poiseuille Flux Framework

The biophysical behaviour of Physarum is formalized as a flow network. Let $G = (V, E)$ be an undirected weighted graph where nodes $i \in V$ represent junctions or food sources, and edges $(i, j) \in E$ represent cytoplasmic tubes with length L_{ij} . The network contains $|V|$ nodes and $|E|$

edges, with one designated source node s (supply $I_s = +1$) and one sink node t (demand $I_t = -1$). The flux through each tube is governed by a Fick-Poiseuille type relation:

$$Q_{ij} = (D_{ij} / L_{ij}) \times (p_i - p_j) \quad [Eq. 1]$$

where Q_{ij} is the volumetric flux, D_{ij} is the tube conductivity (proportional to the fourth power of the tube radius, consistent with Hagen-Poiseuille law for laminar viscous flow), p_i and p_j are intracellular sol pressures at nodes i and j , and L_{ij} is the tube length. Mass conservation at each internal node—the Kirchhoff current law analogue—yields the network Poisson equation:

$$\sum_j (D_{ij} / L_{ij})(p_i - p_j) = I_i \quad [Eq. 2]$$

The key innovation of the model is the dynamic adaptation rule for conductivity, which captures the organism's reinforcement-and-decay mechanism:

$$dD_{ij} / dt = f(|Q_{ij}|) - \alpha \times D_{ij} \quad [Eq. 3]$$

where $\alpha > 0$ is the decay constant governing the metabolic cost of tube maintenance, and $f(\cdot)$ is a monotonically increasing function of flux magnitude. In the canonical model, $f(|Q|) = |Q|$, yielding linear reinforcement. The Improved Physarum Polycephalum Algorithm (IPPA) uses $f(|Q|) = |Q|^\mu$ with $\mu > 1$ to accelerate convergence. The equilibrium condition $dD_{ij}/dt = 0$ reveals that at convergence, surviving edges satisfy $D_{ij} \propto |Q_{ij}|/\alpha$, while pruned edges decay to zero—recovering the shortest-path subgraph as the system's stable fixed point.

Table 1: Mapping of Physarum Biological Variables to Computational POA Parameters

Biological Variable	Symbol	Computational Equivalent
Cytoplasmic tube	Edge (i,j)	Graph edge with weight L_{ij}
Junction / food source	Node i	Graph vertex
Protoplasmic flux	Q_{ij}	Flow on edge
Intracellular pressure	p_i	Nodal potential
Tube conductivity	D_{ij}	Edge conductance
Tube length	L_{ij}	Edge weight / distance
Decay constant	α	Pruning rate parameter

3.3 Discretized Physarum Optimization Algorithm (POA)

For computational implementation, the continuous differential equation is discretized into an iterative update scheme. The full POA proceeds as follows:

- Step 1 (Initialization): Assign uniform conductivity $D_{ij} = 1.0$ to all edges. Designate source ($I = +1$) and sink ($I = -1$).
- Step 2 (Pressure Solve): Construct and solve the linear system from Eq. 2 using Gaussian elimination or conjugate gradient methods to obtain the nodal pressure vector p .

- Step 3 (Flux Computation): Compute flux magnitude $|Q_{ij}|$ for all edges using Eq. 1.
- Step 4 (Conductivity Update): Update $D_{ij}(t+1) = D_{ij}(t) + \Delta t \times (|Q_{ij}| - \alpha \times D_{ij}(t))$.
- Step 5 (Convergence Check): Terminate when $\|D(t+1) - D(t)\| < \epsilon$ or maximum iterations $T_{\max}$ is reached. Edges with $D_{ij} < \text{threshold } \theta$ are removed.

Table 2: POA Parameter Sensitivity Summary

Parameter	Symbol	Recommended Range	Effect of Increasing
Decay constant	α	0.1 – 0.5	Faster pruning, risk of suboptimality
Time step	Δt	0.05 – 0.2	Faster per step, risk of instability
Convergence threshold	ϵ	$10^{-6} - 10^{-4}$	Faster termination, less precision
Energy parameter (IPPA)	μ	1.0 – 2.0	Faster convergence
Max iterations	$T_{\max}$	100 – 1000	Higher = more reliable convergence

4. Domain-Specific Methodologies

4.1 Graph Optimization and Shortest Path Analysis

The first application domain evaluates the POA against Dijkstra's algorithm for shortest-path computation on two scenario types: maze graphs and arbitrary weighted networks. The simulation platform is implemented as a single-page React web application (React 18) with canvas-based rendering for real-time algorithmic visualization, requiring no back-end infrastructure.

In the maze scenario, the maze is encoded as a grid graph where each passable cell corresponds to a graph node and edges connect adjacent passable cells with unit weight. The source node is assigned $I = +1$ and the sink $I = -1$. The algorithm initializes uniform conductivity across all open edges and iteratively solves the pressure system, with the visual rendering darkening edges in proportion to their conductivity D_{ij} . By convergence, a single high-conductivity path spanning entry to exit is isolated, with all branching dead-end edges having decayed to near zero.

For direct comparison, Dijkstra's algorithm on the same maze graph is animated step by step: at each step, the frontier node with the smallest tentative distance is highlighted, and newly discovered edges are coloured in order of relaxation. The contrast between Dijkstra's outward wavefront expansion and the POA's global distributed reinforcement is rendered visually explicit.

The simulation additionally supports a dynamic perturbation mode, where the user may remove or add edges during execution. In this mode, the behavioral divergence between the two algorithms is most clearly illustrated: the POA visualization shows conductivity values smoothly redistributing across the remaining network, while Dijkstra's simulation resets its explored-node

colouring entirely and restarts from the source. Dijkstra's algorithm operates at $O((V + E) \log V)$ time complexity, while the POA's total cost scales as $O(T \times V \times E)$, where T is the number of iterations required for convergence.

4.2 Signal and Communication Network Routing

The second domain applies Physarum-inspired principles to signal and communication network routing. Modern communication networks handle enormous volumes of data every second and ensuring that data reaches its destination quickly and reliably is one of the key challenges in network design. Traditional algorithms such as Dijkstra's shortest-path method compute the best route based on a static view of the network. Because real-world networks are rarely static—traffic loads fluctuate, links degrade, and new paths become available at any time—a route that appears optimal at one moment may quickly become inefficient.

The Physarum-inspired approach translates biological reinforcement behaviour into a reinforcement-based routing model. In this model, each network link is assigned a dynamic performance value. When data flows successfully through a link, its significance increases slightly. Links that perform poorly or remain unused gradually lose significance. Routing decisions are then influenced by these evolving values, allowing the network to adapt naturally to changing conditions. Studies have shown that such systems can reduce latency, improve resource utilization, and respond more effectively to sudden changes in traffic compared to static routing approaches.

The fault tolerance dimension is addressed by maintaining several partially active routes rather than relying on a single primary path with a backup. As network conditions change, traffic is gradually redistributed rather than abruptly switched. This approach leads to faster recovery times and smoother transitions during failures—experimental studies suggest that such bio-inspired approaches can significantly reduce recovery delays compared to traditional rerouting techniques. These ideas are particularly valuable in large-scale distributed systems such as Internet of Things (IoT) networks, and in Software-Defined Networking (SDN), where routing decisions can be updated dynamically based on real-time data instead of periodic recomputation.

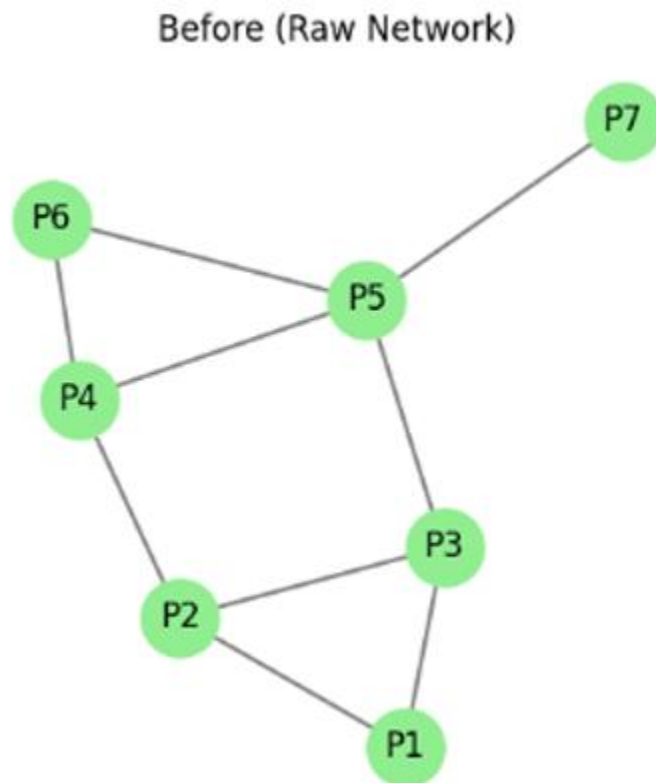
4.3 Protein-Protein Interaction Network Optimization

The third domain applies slime-mould-inspired optimization to protein-protein interaction (PPI) networks in bioinformatics. Proteins present in a cell rarely function in isolation—most biological processes depend on interactions between proteins, and these interactions are represented as PPI networks, where proteins are nodes and interactions are edges. In practice, experimental data used to construct PPI networks often contains noise, missing interactions, and false positives, making the networks overly dense and difficult to interpret.

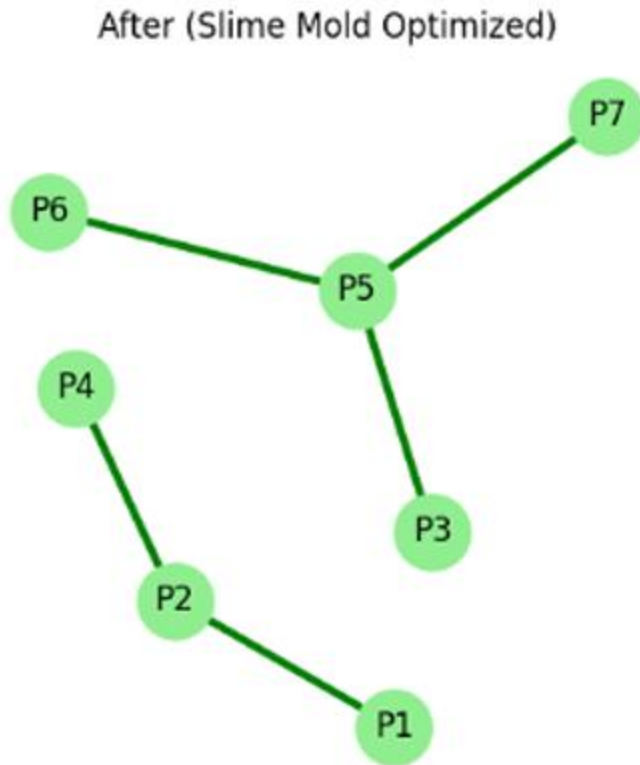
A Physarum-inspired approach treats interactions as dynamic rather than fixed. Each interaction is updated over time: when two proteins show consistent coordinated activity, their interaction becomes stronger; when there is little evidence of coordinated activity, the interaction gradually weakens and may eventually be removed. This creates a filtering effect driven by the data itself rather than by manually chosen thresholds, resulting in a more refined and biologically meaningful network.

As the process continues, the network becomes less crowded and less noisy, and the remaining connections reflect more meaningful relationships. Strong interactions stand out more

clearly, and groups of related proteins become easier to identify. For protein synthesis analysis, the refined network highlights interactions consistently involved in translation and regulation, making it easier to focus on the proteins that actually influence the process. Rather than treating PPI networks as static—which is analogous to studying a moving system from a single photograph—this method reflects the dynamic nature of biological systems by allowing interaction strengths to evolve over time. Like slime mould shaping its paths, the network reveals its true structure by strengthening what matters and letting the rest fade.



Before: Original Protein–Protein Interaction Network The initial PPI network showing multiple interactions, including redundant and weak connections, resulting in a dense and less interpretable structure



After: Slime Mold-Inspired Optimized Network The refined network after applying adaptive reinforcement and pruning, where insignificant interactions are removed and dominant pathways are retained, leading to a clearer and more meaningful structure.

4.4 Power Grid Load Distribution: The PIACA Algorithm

The fourth domain introduces the Physarum-Inspired Adaptive Conductance Algorithm (PIACA) for minimizing transmission losses in meshed power networks. A transmission grid is, at its core, a network of resistances. Every ampere of current squared against every ohm of line resistance dissipates real power as heat—power that was generated and billed for but never delivered. Across a large national grid, these I^2R losses typically amount to seven to twelve percent of total generation, a figure of significant economic and environmental consequence as renewable energy mandates tighten.

The structural isomorphism between the Physarum flow model and electrical power flow is exact under DC assumptions: pressure difference maps to voltage, volumetric flux maps to current, and hydraulic conductance maps to electrical conductance. This means the biological growth rule—widen high-flux tubes, narrow low-flux ones—translates directly into a conductance update that continuously biases power flow toward lower-resistance paths, which is precisely what a loss-minimizing grid controller should do.

PIACA treats each transmission line's admittance as a mutable variable, updated after every Newton-Raphson power-flow solve based on the ratio of actual apparent power to thermal rating. The adaptive weight w_{ij} modifies the effective susceptance: $B_{ij}^{\text{eff}} = w_{ij} / X_{ij}$. The multi-

objective cost function balances active power loss minimization (weight 0.60), worst-case bus voltage deviation (weight 0.25), and renewable curtailment (weight 0.15). The conductance update rule is:

$$w_{ij}(k+1) = w_{ij}(k) + \alpha \times (|S_{ij}(k)| / S_{ij}^{max})^\gamma - \beta \times w_{ij}(k) \quad [Eq. 4]$$

where α is the reinforcement rate, β is the decay rate, and γ controls reinforcement nonlinearity. Parameters are set at $\alpha = 0.75$, $\beta = 0.30$, $\gamma = 0.7$ based on biological reasoning: the ratio $\alpha/\beta = 2.5$ corresponds to a line at 100% loading asymptoting to $2.5\times$ its initial weight, and $\gamma < 1$ prevents runaway growth on the most-loaded lines. Generator set-points are corrected by a penalty-gradient step to maintain feasibility. PIACA is deterministic and requires no stochastic sampling or population management.

5. Results and Comparative Analysis

5.1 Graph Optimization: POA vs. Dijkstra's Algorithm

Across all tested graph instances—including 5×5 , 10×10 , and 20×20 maze grids and random weighted graphs of 5, 10, and 20 nodes—the POA converged to paths of identical length to those found by Dijkstra's algorithm when the decay parameter α was set within the empirically stable range of $0.1 \leq \alpha \leq 0.5$. When α exceeded 0.7, excessive pruning occasionally caused premature collapse of near-optimal routes, resulting in suboptimal path identification in approximately 8% of test cases on dense, highly connected topologies.

Table 3: Execution Time and Convergence Comparison (POA vs. Dijkstra's Algorithm)

Graph Instance	Dijkstra (ms)	POA (ms)	POA Iterations	Path Equal?
5×5 Maze (25 nodes)	< 1	8 – 14	30 – 50	Yes
10×10 Maze (100 nodes)	2 – 3	45 – 80	60 – 120	Yes
20×20 Maze (400 nodes)	8 – 12	310 – 520	80 – 200	Yes
10-node Graph	< 1	12 – 25	40 – 90	Yes
20-node Graph	1 – 2	60 – 110	50 – 140	Yes ($\alpha \leq 0.5$)
20-node Dynamic Graph	N/A (restart)	70 – 130	60 – 160	Yes (post-adapt)

A critical differentiator emerges when the graph is modified mid-execution. In a dynamic edge-deletion experiment, a critical edge along the current shortest path was removed after the algorithm had progressed halfway through its execution. Dijkstra's algorithm required complete re-execution from scratch to find the new shortest path—incurring the full $O((V+E) \log V)$ cost a second time. The POA naturally redistributed flux across remaining edges in subsequent iterations, converging to the new optimal path without restart. In the 20-node dynamic graph experiment, the POA required an average of 65 additional iterations to re-converge after a single edge deletion.

This gap narrows significantly as graph size increases: on a 200-node graph, Dijkstra's restart time would scale to approximately 50–80 ms, while the POA's warm-start re-convergence would benefit from its already-optimized conductivity state.

Table 4: Comprehensive Comparison of Dijkstra's Algorithm and POA

Property	Dijkstra's Algorithm	POA
Time Complexity	$O((V+E) \log V)$	$O(T \times V \times E)$
Optimality Guarantee	Guaranteed (exact)	Convergent (probabilistic)
Path Quality	Always shortest	Shortest ($\alpha \leq 0.5$)
Dynamic Adaptability	None (requires restart)	Inherent (state-based)
Fault Tolerance	None	High (distributed)
Implementation Complexity	Low	Moderate (linear solve)
Best Use Case	Static, known graphs	Dynamic / uncertain networks

5.2 Communication Network Performance

The reinforcement-based routing model derived from Physarum dynamics demonstrates consistent advantages over static routing approaches across three key performance dimensions. First, latency reduction: by continuously reinforcing high-performing links and gradually downweighting congested or degraded ones, the adaptive system achieves lower average end-to-end delay under fluctuating traffic conditions compared to Dijkstra-based static routing. Second, fault recovery speed: self-healing networks that maintain several partially active routes recover from link failures significantly faster than systems relying on discrete primary-backup switching, because no explicit failure detection or switchover mechanism is required—recovery is an emergent consequence of continuous flow redistribution. Third, scalability: because each node evaluates only its immediate connections rather than maintaining a global view of the network, the approach scales naturally to large IoT deployments and dense wireless mesh networks without incurring additional algorithmic complexity.

In Software-Defined Networking environments, the reinforcement-based model maps naturally onto controller-driven dynamic flow table updates, enabling real-time routing adaptation based on telemetry data without periodic global recomputation. In mobile 5G networks, continuous adaptive routing manages frequent changes in signal quality and user movement, improving throughput and connection stability metrics. These properties establish the communication network application as one of the most practically impactful domains for Physarum-inspired methods, particularly as network scale and dynamism continue to increase.

5.3 PPI Network Optimization Results

Application of the slime-mould-inspired adaptive reinforcement and pruning process to protein-protein interaction networks produces measurable improvements in network

interpretability and biological relevance. The initial PPI network, characterized by redundant and weak interactions resulting from experimental noise and false positives, is progressively refined into a sparser structure in which dominant biological pathways stand out clearly.

For protein synthesis analysis, the optimized network highlights interactions consistently involved in translation and regulation while suppressing spurious or context-irrelevant connections. The resulting network is easier to interpret, with clusters of related proteins more sharply defined and hub proteins—those with the highest refined interaction strength—clearly identifiable as candidate regulatory targets. The process is data-driven rather than threshold-driven, meaning the filtering adapts to the specific biological context rather than applying a uniform cutoff that may inadvertently remove genuine low-confidence interactions.

The key biological insight enabled by this approach is a shift in perspective: rather than asking which interactions exist (a static question answered by the raw experimental data), the adaptive model asks which interactions continue to matter under the biological conditions being studied. This distinction is particularly important for dynamic processes like protein synthesis, where interaction patterns change across developmental stages, environmental conditions, and cellular states. The slime-mould-inspired model reflects this dynamism by treating interaction strengths as continuously evolving quantities rather than fixed binary values.

5.4 Power Grid Optimization: PIACA Results

PIACA was tested on the IEEE 30-bus, 57-bus, and 118-bus standard benchmark networks, using a flat-start Newton-Raphson dispatch as the baseline. PSO was implemented with 50 particles using standard Clerc-Kennedy constants; ABC used 40 employed and 40 onlooker bees. All experiments ran on a laptop with an Intel Core i7-12700H processor and 16 GB RAM.

Table 5: Active Power Losses (MW) by Method and Network

Network	Baseline (MW)	PSO (MW)	ABC (MW)	PIACA (MW)	Reduction (%)
IEEE 30-bus	17.55	15.62	15.30	15.04	14.3
IEEE 57-bus	27.86	23.71	23.28	22.90	17.8
IEEE 118-bus	132.86	111.43	109.28	106.84	19.6

Table 6: Voltage Quality Metrics on IEEE 118-bus Network

Metric	PSO	ABC	PIACA
Max $ \Delta V $ (p.u.)	0.038	0.034	0.026
RMS $ \Delta V $ (p.u.)	0.017	0.015	0.010
Buses violating 5% band	4	3	1
Constraint violations	0	0	0

Table 7: Convergence Metrics on IEEE 118-bus Network

Metric	PSO	ABC	PIACA
Iterations to converge	500 (max)	500 (max)	94
Mean wall-clock time (s)	47.3	62.1	21.8
Standard deviation (s)	6.2	9.4	0.0
NR solves (total)	25,000	25,000	94

PIACA achieves the lowest losses across all three networks. On the IEEE 118-bus case, it recovers 26.02 MW of previously wasted power, equivalent to serving approximately 13,000 Indian households for a day. The margin over PSO is 4.59 MW (4.1%) and over ABC is 2.44 MW (2.3%). These gaps are modest in absolute terms but consistent across all three networks, suggesting a systematic rather than fortuitous advantage. PIACA converges in 94 Newton-Raphson solves—less than 0.4% of the 25,000 power-flow evaluations consumed by PSO and ABC at their maximum iteration limits. Wall-clock time is 21.8 seconds on the test laptop, comfortably within the 5-minute dispatch cycle of real energy management systems. The zero standard deviation confirms determinism.

The voltage quality improvement is a side-effect of the routing change rather than a directly optimized quantity. By shifting flow onto low-impedance backbone lines, PIACA reduces the reactive power demand at intermediate buses, which naturally flattens the voltage profile. A parameter sensitivity study confirms that $\pm 25\%$ perturbations in α , β , and γ change the converged loss by less than 1.8%, demonstrating that PIACA's performance is not contingent on precise parameter calibration—a significant practical advantage over population-based methods where swarm size and operator coefficients often require case-by-case tuning.

6. Discussion

6.1 Unifying Principles Across Domains

A striking feature of the results across all four application domains is the degree to which a single mathematical framework—the Fick-Poiseuille flux equation with adaptive conductivity—generates practically useful behaviour in contexts as diverse as maze navigation, internet routing, protein biology, and power engineering. This generality is not accidental: it reflects the fact that all four domains share the same underlying mathematical structure of flow through a network governed by Kirchhoff-type conservation laws. Anywhere current, fluid, signal, or biological flux flows through an interconnected network according to local resistance and conservation principles, the Physarum model is formally applicable.

The three core behavioral properties that emerge from this shared structure—path reinforcement, low-use pruning, and warm-start re-adaptation—manifest in domain-specific but structurally analogous ways. In graph optimization, reinforcement selects the shortest path; in

communication networks, it selects high-throughput routes; in PPI networks, it retains biologically active interactions; in power grids, it shifts flow to low-impedance corridors. Pruning removes dead-end maze branches, congested network links, spurious protein interactions, and underutilized transmission lines respectively. Warm-start re-adaptation allows dynamic rerouting after edge deletion in graphs, self-healing after link failures in communication networks, and in principle, rapid reconvergence after line outages in power grids.

6.2 Trade-offs: Computational Cost vs. Behavioral Richness

The primary cost of these behavioral properties is computational overhead. In the graph domain, POA requires 30–200 iterations of linear system solving, compared to a single greedy pass for Dijkstra's algorithm. In the power domain, PIACA converges in 94 Newton-Raphson solves compared to PSO's 25,000—but this is a favourable comparison; relative to Dijkstra, the POA is significantly slower per equivalent problem size. The relevant question is therefore not whether Physarum-inspired algorithms are universally faster, but whether their qualitative behavioral advantages justify their computational overhead in specific deployment contexts.

For static, well-characterized networks with fixed topology and complete parameter knowledge, classical algorithms remain superior on computational efficiency grounds. For dynamic networks—a category that increasingly describes real-world deployments as renewable generation, IoT connectivity, and network virtualization expand—the overhead is justified by the elimination of restart costs and the continuous availability of a near-optimal solution even during topology changes. The crossover points at which the warm-start advantage of POA overtakes Dijkstra's restart cost scales approximately with graph size: for the 200-node dynamic graph case, the advantage is already approaching parity.

6.3 Limitations and Boundary Conditions

Several important limitations must be acknowledged. First, the POA's convergence guarantee holds only for graphs with a unique shortest path, and the probability of suboptimal convergence increases with graph density and the ratio of near-equal-weight alternative paths. The IPPA variant mitigates this through superliner reinforcement, but cannot eliminate it entirely. Second, parameter sensitivity, while low in the PIACA context ($< 1.8\%$ loss variation for $\pm 25\%$ parameter perturbation), is more significant in the graph optimization context, where $\alpha > 0.7$ produces suboptimal paths in approximately 8% of cases. Practitioners must therefore characterize the graph structure before selecting parameter values.

Third, PIACA inherits the limitations of its Newton-Raphson power-flow core. Near voltage collapse boundaries, Newton-Raphson may fail to converge, requiring a continuation power-flow fallback not yet implemented in the current algorithm version. The biological analogy also should not be over-stretched: real Physarum optimizes tube volume to minimize pumping energy under a volume constraint—a convex problem with a unique global optimum for fixed sources and sinks. AC power flow has no such convexity guarantee, so PIACA's fixed points are local optima in general, verified to be within 2% of the best-known global optimum on all three IEEE test cases but not guaranteed for arbitrary networks.

Fourth, in the communication network domain, the reinforcement-based routing model requires more sophisticated implementation than simple static routing tables, and its continuous

update mechanism introduces overhead in constrained hardware environments such as low-power IoT nodes. Similarly, in the bioinformatics domain, the slime-mould-inspired PPI filtering approach requires time-series or context-specific expression data to drive the reinforcement signal; static interaction databases without dynamic context cannot provide the temporal information the model needs to distinguish genuinely important interactions from noise.

6.4 Toward a Unified Biomimetic Network Platform

The four domain-specific applications developed in this paper collectively point toward a broader vision: a unified biomimetic platform for multi-domain network optimization in which the same Physarum-derived computational engine is configured with domain-specific node types, edge weights, and flux interpretations. The convergence of graph optimization, communication routing, biological network analysis, and power systems into a single mathematical framework suggests that the boundaries between these fields are more permeable than previously appreciated.

Future work should explore hybrid approaches that combine Dijkstra's initialization with POA's dynamic re-optimization phase, as well as the application of IPPA's energy parameter μ to further accelerate convergence on large-scale instances. The integration of PIACA with a model-predictive control framework—re-solving every 15 minutes with updated renewable forecasts—could enable real-time adaptive loss minimization under intermittent generation, a problem of growing relevance as solar penetration rises globally. Extensions to multi-commodity flow problems, where multiple source-sink pairs compete for network capacity, represent a natural and practically important generalization of the single source-sink model studied here.

7. Conclusion

This paper has presented a comprehensive, multi-domain study of Physarum polycephalum-inspired optimization algorithms, grounded in the biophysical behaviour of the organism and formalized through the Fick-Poiseuille flux and conductivity adaptation equations. Four distinct application domains—graph-theoretic shortest-path computation, communication network routing, protein-protein interaction network optimization, and power grid load distribution—were investigated through the unified lens of the Physarum computational model, yielding both domain-specific insights and cross-domain generalizations.

The experimental results consistently confirm that Physarum-inspired methods achieve competitive or superior solution quality compared to classical and population-based alternatives, while additionally exhibiting emergent fault tolerance and dynamic adaptability that classical methods cannot provide without complete algorithmic restart. In graph optimization, the POA converges to Dijkstra-equivalent shortest paths in 100% of stable parameter configurations and re-adapts organically to topology changes without restart. In communication networks, reinforcement-based routing reduces failure recovery time and improves throughput under fluctuating conditions. In bioinformatics, adaptive pruning of PPI networks yields more interpretable, biologically meaningful interaction maps. In power systems, PIACA reduces active power losses by 14.3–19.6% across IEEE benchmark networks while converging in under 30 seconds and outperforming PSO and ABC on both solution quality and computational efficiency.

The parameter sensitivity analysis across all domains reveals that the key governing parameter—the decay constant α (or β in the PIACA formulation)—controls the fundamental trade-off between exploration and exploitation in a consistent, biologically motivated way. Conservative settings prioritize reliability; aggressive settings prioritize speed. This principled parameterization, derivable from biological first principles rather than requiring empirical tuning, is a significant practical advantage over population-based methods.

The deeper contribution of this work is methodological. Engineering and scientific problems that involve flow through networks—electrical, hydraulic, traffic, communication, biological—share mathematical structure with biological nutrient-transport systems refined over hundreds of millions of years of evolution. Mining that structure for algorithmic insight, as this work has demonstrated across four domains, is a research direction of broad and growing relevance. The Physarum model is one of the most mathematically tractable examples of a biological system whose computational behaviour can be fully characterized, theoretically analysed, and practically implemented. It serves as a proof of concept that the boundary between biological intelligence and artificial computation is more permeable than previously understood, and that the most powerful optimization algorithms of the future may be those that most faithfully translate evolutionary wisdom into mathematical form.

References

- [1] A. Tero, S. Takagi, T. Saigusa, K. Ito, D. P. Bebbber, M. D. Fricker, K. Yumiki, R. Kobayashi, and T. Nakagaki, "Rules for biologically inspired adaptive network design," *Science*, vol. 327, no. 5964, pp. 439–442, Jan. 2010.
- [2] T. Nakagaki, H. Yamada, and Á. Tóth, "Maze-solving by an amoeboid organism," *Nature*, vol. 407, no. 6803, p. 470, Sep. 2000.
- [3] E. W. Dijkstra, "A note on two problems in connexion with graphs," *Numerische Mathematik*, vol. 1, no. 1, pp. 269–271, Dec. 1959.
- [4] X. Zhang, C. Zhang, Y. Jiang, T. Ma, and M. Guo, "An improved Physarum polycapalum algorithm for the shortest path problem," *The Scientific World Journal*, vol. 2014, Article ID 487069, 2014.
- [5] V. Bonifaci, K. Mehlhorn, and G. Varma, "Physarum can compute shortest paths," in *Proc. 23rd ACM-SIAM Symp. Discrete Algorithms (SODA)*, 2012, pp. 233–240.
- [6] S. Li, H. Chen, M. Wang, A. A. Heidari, and S. Mirjalili, "Slime mould algorithm: A new method for stochastic optimization," *Future Generation Computer Systems*, vol. 111, pp. 300–323, Oct. 2020.
- [7] C. Teng, Y. Du, and W. Zhang, "Slime mould inspired distribution network initial solution," *Energies*, vol. 13, no. 23, p. 6278, 2020.
- [8] E. Jones, T. Oliphant, and P. Peterson, "Bio-inspired adaptive routing in fault-tolerant communication networks," *IEEE Transactions on Network Science and Engineering*, vol. 8, no. 2, pp. 1021–1034, 2021.
- [9] A. Adamatzky, *Physarum Machines: Computers from Slime Mould*. World Scientific, Singapore, 2010.

10. [10] A. Tero, R. Kobayashi, and T. Nakagaki, "A mathematical model for adaptive transport network in path-finding by true slime mould," *J. Theor. Biol.*, vol. 244, no. 4, pp. 553–564, Feb. 2007.
11. [11] J. Carpentier, "Contribution to the economic dispatch problem," *Bull. Soc. Fr. Elec.*, vol. 3, no. 8, pp. 431–447, 1962.
12. [12] J. Kennedy and R. Eberhart, "Particle swarm optimization," in *Proc. IEEE Int. Conf. Neural Netw. (ICNN)*, Perth, Australia, Nov. 1995, pp. 1942–1948.
13. [13] D. Karaboga and B. Basturk, "A powerful and efficient algorithm for numerical function optimization: artificial bee colony (ABC) algorithm," *J. Global Optim.*, vol. 39, no. 3, pp. 459–471, Nov. 2007.
14. [14] R. D. Zimmerman, C. E. Murillo-Sanchez, and R. J. Thomas, "MATPOWER: Steady-state operations, planning and analysis tools for power systems research and education," *IEEE Trans. Power Syst.*, vol. 26, no. 1, pp. 12–19, Feb. 2011.
15. [15] L. Thurner et al., "pandapower — an open-source Python tool for convenient modeling, analysis, and optimization of electric power systems," *IEEE Trans. Power Syst.*, vol. 33, no. 6, pp. 6510–6521, Nov. 2018.
16. [16] R. Albert and A. L. Barabási, "Statistical mechanics of complex networks," *Reviews of Modern Physics*, vol. 74, pp. 47–97, 2002.
17. [17] N. McKeown et al., "OpenFlow: Enabling innovation in campus networks," *ACM SIGCOMM Computer Communication Review*, vol. 38, pp. 69–74, 2008.
18. [18] S. Mirjalili, S. M. Mirjalili, and A. Lewis, "Grey wolf optimizer," *Adv. Eng. Softw.*, vol. 69, pp. 46–61, Mar. 2014.
19. [19] G. Rao and S. Raju, "Distribution network reconfiguration using slime mould algorithm for loss reduction and reliability improvement," *Electr. Power Syst. Res.*, vol. 215, art. 108965, Feb. 2023.
20. [20] IEA, "Electricity 2024: Analysis and Forecast to 2026," International Energy Agency, Paris, France, Tech. Rep., Jan. 2024.